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An analytical model from physical parameters to minimum ranging time for photon-counting LiDARs

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ABSTRACT

Long-distance light detection and ranging (LiDAR) has been highly demanded for applications on unmanned vehicles and drones. CMOS-fabricated single-photon avalanche diodes (SPADs) play a key role in the receiver end due to their high photo-sensitivity and readiness for system-on-chip integration. However, the large amounts of involved components together with the diverse ranging conditions make engineering and optimizing these modules a daunting challenge. In this work, we have developed an analytical model for calculating minimum ranging time from the physical parameters for a photon-counting LiDAR. The experimental verifications of the model have been performed and a good consistency has been obtained. Our work enables architecture design and optimization for making low-cost high-performance SPAD LiDARs.

1. Introduction

Light detection and ranging (LiDAR) system have attracted increasing attentions for its potential for perception layer in autonomous driving and advanced driving assistance system [1–14]. Among various sensors on vehicles, LiDARs advantages high imaging resolution and reliable ranging capability which is crucial for road safety requirements. In the past decade, inspired by the pioneer work by Niclass and his co-workers [1,2], using complementary metal–oxide–semiconductor (CMOS) silicon-based single-photon detectors, named single-photon avalanche diodes (SPADs) has been widely investigated for LiDAR applications [15]. CMOS SPADs not only largely reduce the required laser power but also dramatically simplify the accompanying circuit design due to their inherent digital nature and easy integration with subsequent signal processing circuits on the same chip. In addition, the excellent timing resolution down to picosecond range makes SPADs nearly perfect for pulsed-mode direct time-of-flight (ToF) ranging or photon-counting based LiDAR [15].

In the aspect of imaging method, the present approaches are quite diverse in beam steering scheme, including pixel-by-pixel (two-axis) scanning, line (one-axis) or column scanning, and flash imaging [1–14]. This diversity together with the massive physical parameters involved, including those on the receiver side and on the emitter side, has made analyzing, optimizing, and even benchmarking on LiDAR modules a daunting challenge [3,16–24]. In fact, a longer thought would reveal

that the required time and resource (laser power and chip area) for single-point ranging is the core issue in the system performance. For example, with pixel-by-pixel scanning, the frame rate of the 3-D image is determined by the number of pixels in a frame and single-pixel ranging time. For multiple-pixel imaging scheme, such as line scan or flash imaging, the laser power is shared by those simultaneously ranging pixels but, in return, each pixel is allowed to have a longer ranging time under a fixed frame rate. To perform system engineering in this aspect, we need an efficient method to estimate single-pixel ranging time with given physical parameters in LiDAR modules.

The LiDAR parameters involve internal ones, such as laser source, optics design, sensor characteristics, accompanying circuits, signal processing method, and peak detection algorithm, and external ones, including background sunlight, rain, snow, or fog scatterings, and target reflectivity. To study their effect on module performance, a few literatures on SPAD-based LiDAR modeling have been reported recently. Padmanabhan et al. established a flash-mode LiDAR model starting from the physical parameters [16]. They deployed probabilistic function of incident photons and investigated the effects of spatially coincidence scheme, SPAD grouping detection, and peak detection threshold on detection probability but no comparison with experimental data was presented. Tontini and his coworkers performed a numerical modeling using Monte Carlo method also for flash imaging scheme [17]. The detection rate and ranging precision at short distance (<2 m) as well as ranging histograms were accessed numerically and experimentally with

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good consistence between them. A similar study focused on the histogram resources and the laser power efficiency of flash LiDARs demonstrated the effectiveness of system-level simulation [23]. However, the Monte Carlo simulation is not only time consuming but also not very helpful for understanding the interplay between the physical parameters.

In this paper, we present an analytical LiDAR model and its experimental verifications with two LiDAR setups in dozens of ranging conditions. Our model calculates single-pixel minimum ranging time with the knowledge of physical parameters in a LiDAR module. The laser and sunlight counts with their probabilistic behavior has been addressed and experimentally checked. They are then used to calculate the minimum ranging time with assigned successful detection rate. Without any free-running fitting parameter, an excellent consistency between the calculated and measured minimum ranging times has been obtained. Our work is useful for engineering and optimizing photon-counting-based vehicle LiDARs.

2. Physical Layers of LiDAR setups

Fig. 1 gives the flow of our model from the physical parameters to the calculation of minimum ranging time $T_{\min}$. Starting with the given physical parameters, the previous works [16,17] reported a detailed method to calculate the average laser signal counts triggered by the target back-scattered photons and the sunlight-induced background noise counts, which we respectively define as the return photo-counts per laser pulse at zero SPAD deadtime (denoted as RP_0 , in unit of counts) and the sunlight counts (denoted as C_s , in unit of Hz) hereinafter. In practice, the SPAD deadtime is non-zero so the effective laser counts per pulse, defined as return photo-count per laser pulse or return probability, are denoted as RP and will be calculated from RP_0 , which is the first link (Link 1) to be connected. With given RP and C_s , the counts in timing histogram can be estimated and the minimum ranging time $T_{\min}$ can be calculated accordingly, which is the second link (Link 2) to be established in this work.

Fig. 2 shows the schematics for a simplified ToF module with the laser emitter, the beam steering mechanism, and the receiver. In Fig. 2a, the laser-emitted photons are scattered by the target and return to the receiver through the collection lens in front of the SPAD chip. To calculate the return photo-counts per laser pulse at zero-deadtime RP_0 , we start with the following formula [16,17],

$$RP_0 = N_L \times Rt \times G_f \times T_f \times e_f \times PDE \quad (1)$$

where N_L is the number of emitted photons in a single laser pulse, and Rt is the target reflectivity. The geometrical factor G_f will be discussed

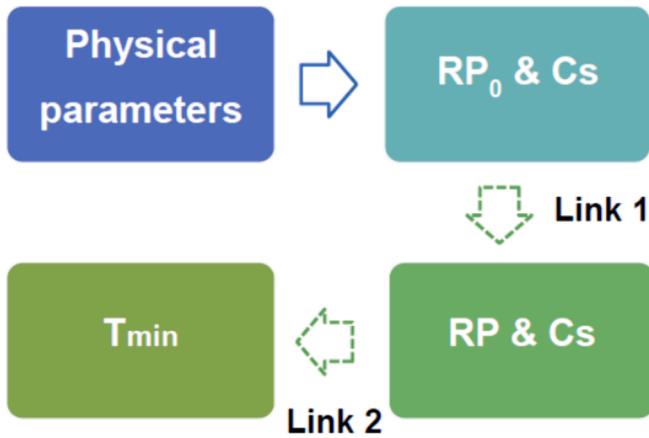


Fig. 1. Schematic flow of our model from the physical parameters of LiDAR module to the predicted minimum ranging time $T_{\min}$. See main text for the definition of the shown symbols and the two links established in this work.

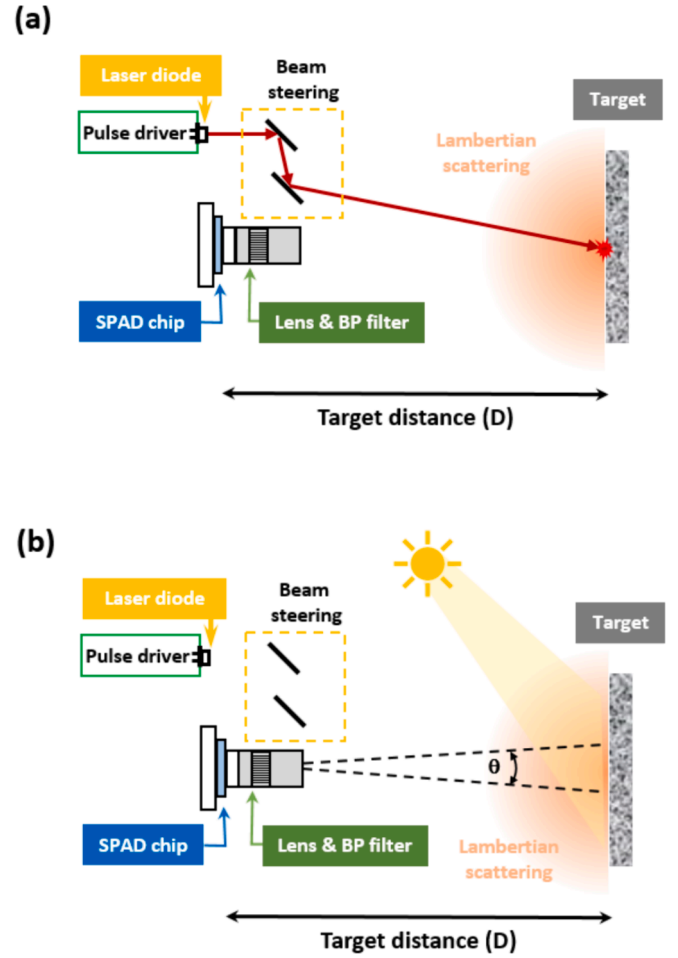


Fig. 2. (a) Schematic for calculating average laser counts per pulse or return photo-count at zero deadtime (RP_0) for a single pixel; (b) Schematic for calculating background counts (C_s) generated by the sunlight in unit time.

shortly. The T_f is the transmissivity of the band-pass filter in front of the receiver. The parameter e_f stands for the focusing efficiency of lens set to account for the imperfect focused laser spot on the SPAD chip. For a laser source with a large divergence angle, the focused spot on the imaging chip could be larger than a single SPAD pixel so the returned photons outside the active SPAD cannot be detected and the e_f will be less than 100 %. At last, the PDE is the photon detection efficiency of the SPAD array, determined by the photon detection probability and array fill factor (FF), that is, $PDE = PDP \times FF$ [15]. In addition, the N_L can be easily calculated by the average laser power P_{ave} , laser repetition rate f_L , and laser wavelength λ as follows.

$$N_L = \frac{P_{ave}}{f_L} \times \frac{1}{1.6 \times 10^{-19}} \times \frac{\lambda(\text{nm})}{1240} \quad (2)$$

Assuming Lambertian scattering, the back-scattered photons arrive at the receiver lens with a geometrical reduction factor G_f due to the relatively small receiving aperture.

$$G_f = \frac{\pi d^2 / 4}{2\pi D^2} \quad (3)$$

where d is the effective aperture diameter of the receiver lens set and D is the target distance. With all the parameters mentioned above, the return photo-counts at zero-deadtime RP_0 can be calculated with Eq. (1)

Referring to Fig. 2b, Eq. (4) is the formula for calculating the sun-generated background noise counts C_s ,

$$C_s = N_s \times (D\theta)^2 \times R_t \times G_f \times \Delta\lambda \times T_f \times PDE \quad (4)$$

where N_s in unit of $/m^2/nm/s$ is the number of sunlight photons per unit area, per unit wavelength, and per unit time at a specific wavelength under a certain sun intensity. The θ is the angle-of-view (AoV) per pixel so $(D\theta)^2$ is the effective area on the target surface for reflecting sunlight. The $\Delta\lambda$ is the band width of the band-pass filter. Assuming the sunlight is uniformly illuminated on the target, the focus efficiency e_f in Eq. (1) is equal to one so it is absent in Eq. (4). The N_s can be calculated with,

$$N_s (/nm/s/m^2) = \frac{P_{sun}}{1.6 \times 10^{-19}} \times \frac{\lambda(nm)}{1240} \quad (5)$$

where P_{sun} is sunlight intensity and can be accessed from [25] and λ is the laser wavelength.

In this work, two setups listed in Table 1 for ranging experiments are used to verify our model. Setup 1 used a 32x32 SPAD array and an edge-emitting laser (EEL) at 905 nm and setup 2 had a 64x128 array and a photonic-crystal surface emitting laser (PCSEL) at 940 nm [26]. The numbers of the two setups are listed in Table 1 and will be detailed in the following.

First, the two sensor chips consisted of 32x32 (chip 1) and 64x128 (chip 2) arrays, respectively, with the same SPAD structure were fabricated with a conventional CMOS 180-nm BCD process. The respective SPAD sizes, array pitches, and fill factor (FF) were 20, μm , 14 μm , and 35 % for chip 1 and 32.5 μm , 26.5 μm and 26.4 % for chip 2. The breakdown voltage of SPAD was ~ 48.0 V. At 53.0 V, the mean dark count rate (DCR) and mean photon detection probability (PDP) were 279 Hz and ~ 10 % at 905 nm for chip 1, and 151 Hz and ~ 5 % at 940 nm for chip 2 [27–30]. The SPAD deadtime of ~ 3.5 ns was realized by deploying high dynamic-range quenching/reset circuits, named as time-gated active reset (TGAR) circuits in our previous report [31]. The on-chip column and row selectors were designed to enable a single SPAD while leaving other SPADs in the off state.

On the other hand, the EEL emitter at 905 nm was a specially designed semiconductor edge-emitting laser [32]. The threshold current and slope efficiency were about 0.3 A and 0.97 W/A, respectively. The PCSEL emitter at 940 nm in TO-can package was purchased from Hamamatsu Ltd. (model No. L13395-04). The two lasers were operated at the pulse repetition rate of 1 MHz with a top-hat-like pulse shape with the full pulse width of about 3.7–4.2 ns and the full-width at half-maximum of about 1.8–2.5 ns. The laser average power used in our both setups were ~ 2.5 mW. Note that the two 10-nm bandwidth of 905/940 nm optical bandpass filters were installed in front of the receiver with the respective transmissivity of 70 % and 90 %. For the 905-nm EEL, the vertical and lateral divergence angles of the laser were about 16–18 degrees and 6.5–8 degrees, respectively. With a single aspheric lens with an aperture, the beam was collimated with the beam divergence angles of about $0.12^\circ \times 0.24^\circ$, which resulted a focus efficiency e_f of ~ 30 % at the target distance of 50 m. In contrast, for the 940-nm

PCSEL, thanks to its symmetrical and small divergence angle of $\sim 1^\circ$, a single focus lens collimated the beam to a divergence angle of $\sim 0.07^\circ$ and resulted an e_f of ~ 91 % at the same distance. For the laser beam steering, a two-axis galvanometer mirror (Edmund, model No. 16041) was installed for pixel optical alignment (and scanning for imaging in other experiments). The receiver lens set had an effective aperture of ~ 12 mm and included the aforementioned optical band-pass filter to suppress the sunlight counts. The laser repetition rate (f_l) was 1 MHz.

In this work, we used a time-correlated single-photon counting card (TCSPEC, Pico-Quant Time-harp 260) as a time-to-digital converter (TDC) to record the timing start signals from the laser driver and the stop events from the SPADs. The time-bin width and the deadtime of the TDC was 250 ps and ~ 2 ns, respectively. To calculate experimental successful detection rate or hit rate, all output signals of the TDC were recorded with the TTTR mode (time-tag time-resolved) of the TCSPEC card for 1 s for subsequent analysis, which was particularly convenient for evaluating minimum ranging time.

Our method for experimentally obtaining minimum ranging time, T_{min} , for single-point ranging under a certain experimental condition is briefed as follows [21]. The center-of-mass (CM) method of 4-ns time window width was used for peak detection and it is worth noting that the chosen 4-ns is close to the SPAD deadtime and the laser pulse width. The distance obtained with the long integration time (1 s in our case) served as the gold standard. If the difference between the detected distance at a certain integration time T and the gold standard was within the CM time window width (4 ns), it was counted as a successful measurement (hit). If not, it was a failed measurement (missing). By doing this for 10000 times from the TTTR data, we can have the detection rate (successful detection) for that integration time T . By repeating this procedure for all integration times, the detection rates as a function of integration time T were obtained. In this work, we define the minimum ranging time T_{min} as the minimum integration time needed to reach a 99 % detection rate.

3. Return photo-counts per pulse – Link 1

Using avalanche breakdown to detect single photons, a SPAD needs accompanying quenching and reset circuits to quench and to recharge it for next photon detection. The process takes typical a few nano-seconds and SPADs cannot detect photons during this period of time, called deadtime. For the scheme of single SPAD for single pixel in this present work, provided that the deadtime is close to the laser pulse width, even there are more than one photo-count could be triggered, only one photo-count can be recorded. This photon “pile-up” effect could be an issue for histogram distortion in ToF ranging [4]. To connect the return photo-counts per pulse at zero deadtime RP_0 with the return photo-count per pulse (at finite deadtime) RP , or named “return probability” by us, the statistical property of RP_0 has to be known. The previous models assumed the values of RP_0 follows a Poisson distribution without experimental verification [16,17]. With this assumption, we can derive the relation between RP and RP_0 ,

$$RP = 100\% - P(k=0) = 1 - e^{-RP_0} \quad (6)$$

where $P(X=k) = (e^{-\lambda} \lambda^k)/k!$ and $\lambda = RP_0$. It is obvious that, for the RP_0 much less than 1, $RP \sim RP_0$ which means the pile-up effect is negligible. For the RP_0 much larger than 1, RP approaches 100 %. Fig. 3a illustrates this relation and, guided by the line of slope = 1, we can see for $RP < 30$ %, RP_0 is very close to RP .

To verify Eq. (6), we extracted the experimental RP from our long integration time (1 s) histogram taken by two setups at various target distances (20–60 m) at low background condition ($C_s < 0.5$ MHz). More specifically, the experimental RP was the total counts in target CM window in measured histogram divided by the number of fired laser pulses (10^6 pulses in 1 s). To extract the experimental RP_0 , the following equation was applied,

Table 1

Physical parameters used for our two LiDAR setups.

	Unit	Setup 1	Setup 2
Laser diode	–	EEL	PCSEL
SPAD array	–	32x32	64x128
λ	nm	905	940
R_t	%	10 or 90	10 or 90
d	mm	12	12
D	m	10–100	10–100
T_f	%	70	90
e_f	%	30	91
PDP@ λ	%	10	5
FF	%	35.0	26.4
P_{ave}	mW	2.5	2.5
P_{sun} @100 klux	W/nm/m ²	0.76	0.44
θ	degree	0.1	0.08
$\Delta\lambda$	nm	10	10

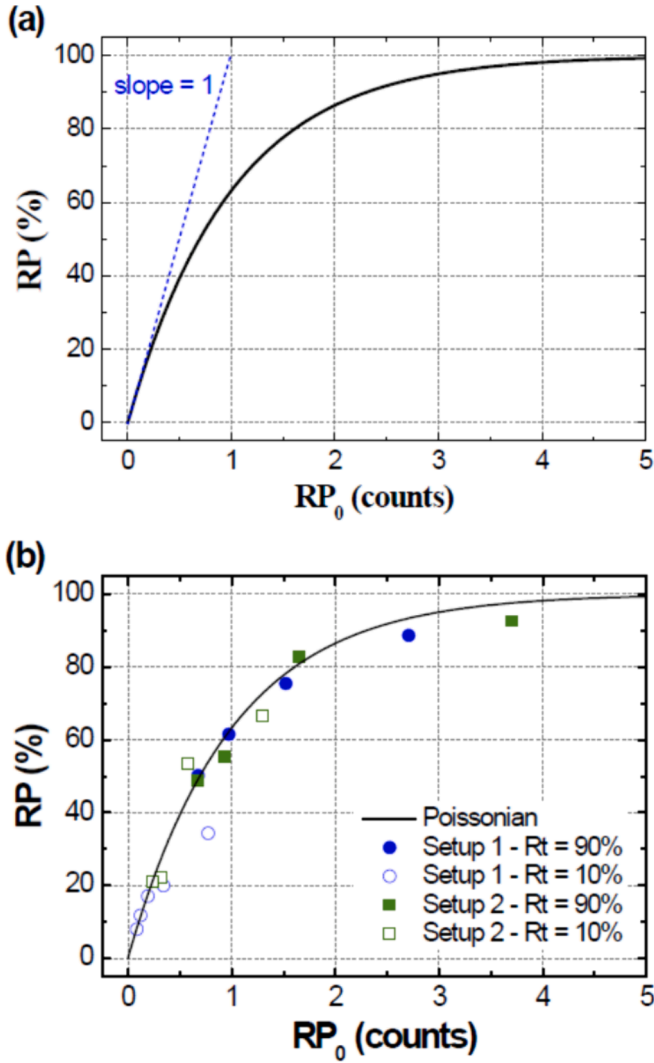


Fig. 3. (a) Theoretical return photo-count RP as a function of return photo-counts at zero deadtime RP_0 ; (b) Experimental RP from two setups for two targets with 10 % and 90 % reflectivities at various target distances. The theoretical curve (solid line) is also plotted for comparison.

$$RP_0(D) = RP_0(\text{ref}) \times \left(\frac{D_{\text{ref}}}{D} \right)^2 \quad (7)$$

where D is the aforementioned target distance. D_{ref} and $RP_0(\text{ref})$ is the longest target distance in the experiment (~ 60 m in setup 1 and ~ 50 m in setup 2). Because the experimental RP at the longest D is the smallest and typically quite small ($< 20\%$), we assumed that its RP_0 was equal to RP and set it as $RP_0(\text{ref})$. The RP_0 of other distances were then calculated with Eq. (7) so the experimental RP as a function of RP_0 was obtained and plotted in Fig. 3b. It is clearly seen that the data points follow Eq. (6) pretty well so we can conclude that the RP_0 is indeed poissonian and Link 1 in Fig. 1 has been established.

At this stage, we have also compared RP_0 in theory and in experiment at various distances with the two setups. Fig. 4 plots the theoretical RP_0 calculated with Eq. (1) and the experimental ones based on the measured RP. For setup 1 in Fig. 4a, the consistency is excellent, particularly for $Rt = 10\%$. For $Rt = 90\%$, the measured RP_0 is slightly higher than the theoretical ones, probably due to unstable emitter (EEL) or non-Lambertian scattering. Fig. 4b, for setup 2 using PCSEL as emitter, shows an acceptable consistency but the data seem more scattered. We reckon that the focus spot size using PCSEL was too small to maintain stability during the measurements. Overall speaking, the

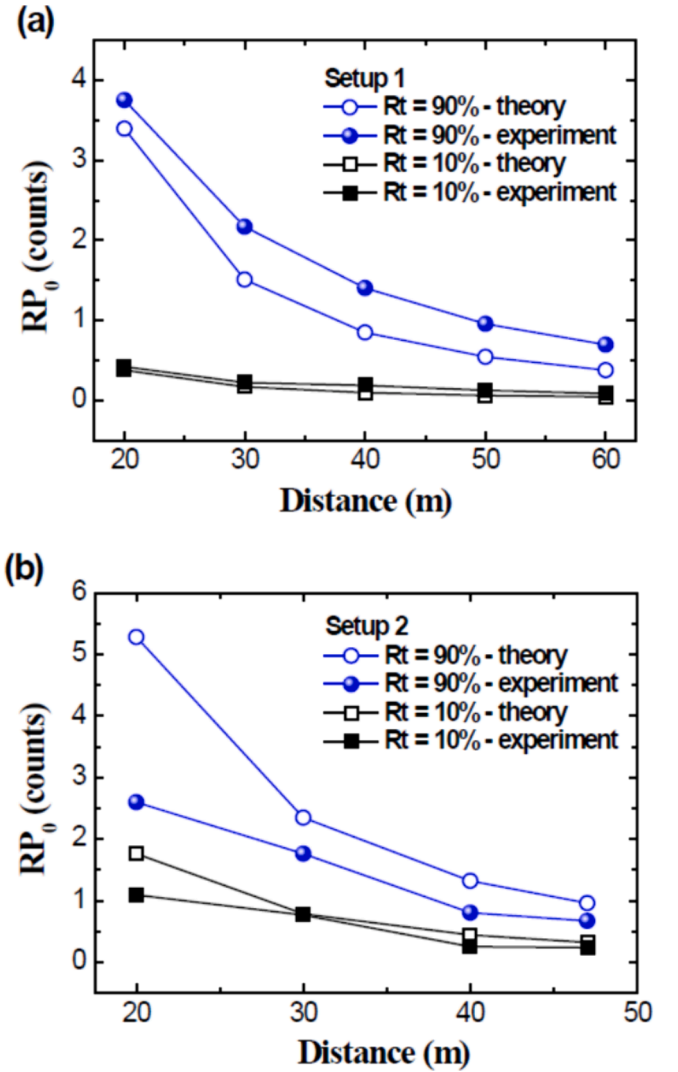


Fig. 4. Theoretical and experimental return photo-counts per pulse at zero deadtime, RP_0 , for the target with 10 % and 90 % reflectivity at various distances under low sunlight background using setup 1 (a), and setup 2 (b).

discrepancies between the theory and the experiment are within 20 % except the single data point of $Rt = 90\%$ at 20 m in Fig. 4b.

In addition, we have also experimentally checked the sunlight background counts C_s calculated by Eq. (4) using the parameters listed in Table 1 and found that they were also quite consistent with the measured ones under several sunlight conditions (not shown here) as the errors were less than 15 %.

4. Minimum ranging time – Link 2

To simulate minimum ranging time T_{min} with known return probability RP and background counts C_s , we have defined two variables, *counts in target window per laser pulse*, denoted as λ_{tw} , and *counts in noise window per laser pulse*, denoted as λ_{nw} within a timing histogram as follows [21]. Fig. 5 shows a collected histogram as an example, λ_{tw} is the total counts in target-located CM window (around 242–246 ns, marked by blue box in Fig. 5) divided by the number of fired laser pulses ('tw' for target window), and λ_{nw} is the noise/background counts in the other CM windows (every window excluding the target window, one of them marked with the black box in Fig. 5, and 'nw' for noise window). Note that, these two variables λ_{tw} and λ_{nw} depend on experimental condition. By using the experimental TTTR data for a single measurement, the

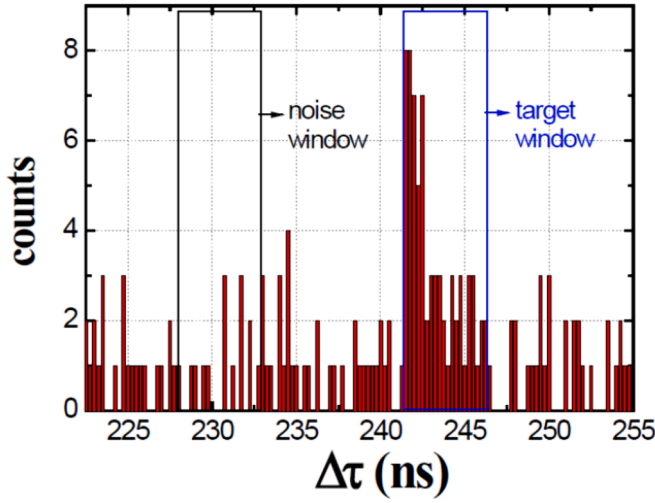


Fig. 5. A ranging histogram as an example for defining the two variables μ_{tw} and μ_{nw} which are counts in target window (marked by the blue box) per laser pulse and counts in noise window (one of them marked by the black box) per laser pulse, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

means and standard deviations of λ_{tw} and λ_{nw} , respectively denoted as μ_{tw} , μ_{nw} , σ_{tw} , and σ_{nw} , can be calculated for all integration times.

We found that, by choosing the CM window approximately equal to the SPAD deadtime, the two variables λ_{tw} and λ_{nw} follow a Gaussian distribution very well, as the example shown in Fig. 6a. Furthermore, by analyzing their standard deviations, we learned that their means and standard deviations follow the relation for a binomial distribution. That is,

$$\sigma = (N_{LP} \times \mu \times (1 - \mu))^{\frac{1}{2}} \quad (8)$$

where the σ stands for either σ_{tw} or σ_{nw} and the μ stands for μ_{tw} and μ_{nw} . N_{LP} is the number of fired laser pulses in the selected integration time T . Note that, because the CM window width is equal to the SPAD deadtime, the counts inside this window size cannot exceed 1 as single SPAD is used. Fig. 6b and c clearly reveals this interesting and important relation, which were obtained with two setups, setup 1 in Fig. 6b and setup 2 in Fig. 6c, with the target at various distance (10–60 m) under four different experimental conditions (cases 1–4) respectively having $R_t = 10\%/90\%$ and high/low sunlight background. It is clear that the two variables in all measurements follow the binomial distribution formula in Eq. (8) very well. We would like to note that the SPAD deadtime (~ 3 ns) being approximately equal to the laser pulse width (~ 4 ns) plays a role in the observed binomial statistics. Accordingly, once the integration time and the means, μ_{tw} and μ_{nw} , are given, the counts inside and outside the target window as well as their probability density functions can be analytically calculated as follows.

For a successful target detection, the total counts in the target-located CM window must be larger than every noise window in the same histogram, as stated in Eq. (9), where HR stands for the hit rate or the detection rate and is set at 99 % for our T_{min} calculation, which gives a miss rate of 1 %. N_{nw} denotes noise window number in each histogram, which was ~ 4000 for the laser repetition rate of 1 M Hz and TDC time-bin width of 250 ps. The first integral on the right-hand side in Eq. (9) is the probability density function of λ_{tw} and the second integral is the cumulative distribution function of λ_{nw} from 0 to corresponding λ_{tw} to the power of N_{nw} because the counts in the target window must be larger than every noise window to be a successful detection.

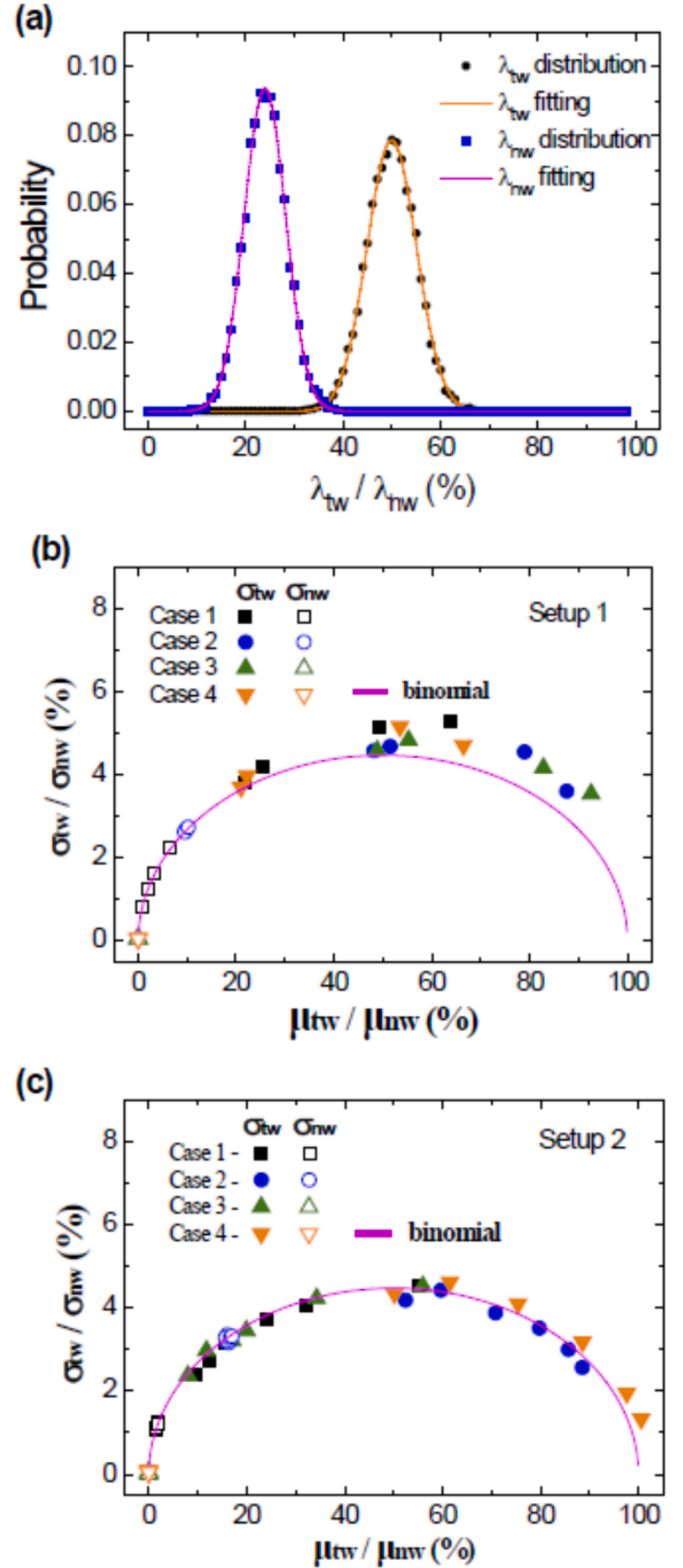


Fig. 6. Measured statistical properties of the two variables λ_{tw} and λ_{nw} . (a) Their distributions and respective Gaussian fittings. (b), (c) Their standard deviations as a function of their mean at the integration time of 80 μ s, together with those for an ideal binomial distribution in (b) for setup 1, and (c) for setup 2 under four ranging conditions: case 1 – $R_t = 10\%$ and high Cs, case 2 – $R_t = 90\%$ and high Cs, case 3 – $R_t = 10\%$ and low Cs, and case 4 – $R_t = 90\%$ and low Cs.

$$HR = \int_{-\infty}^{\infty} \frac{1}{\sigma_{tw}\sqrt{2\pi}} \exp\left(-\frac{(N_{LP}\lambda_{tw} - N_{LP}\mu_{tw})^2}{2\sigma_{tw}^2}\right) \left[\int_{-\infty}^{\lambda_{tw}} \frac{1}{\sigma_{nw}\sqrt{2\pi}} \exp\left(-\frac{(N_{LP}\lambda_{nw} - N_{LP}\mu_{nw})^2}{2\sigma_{nw}^2}\right) d\lambda_{nw} \right]^{N_{mw}} d\lambda_{tw} \quad (9)$$

The theoretical $T_{\min}$ can then be calculated by using Eq. (9) and compared with the experimental ones for all our measurements and the results are plotted in Fig. 7. Clearly, the measured and calculated $T_{\min}$ in the range of a few μs to $\sim 80 \mu\text{s}$ all lay on the line of slope = 1, indicating an excellent agreement between the theory and the experiment.

In addition, we have also performed a simple Monte Carol simulation to check the precision and efficiency of our method. With a given set of μ_{tw} and μ_{nw} , the precision of estimating $T_{\min}$ using our model is comparable with that of the Monte Carol simulation but the used computation time is less than 1 % of that taken by the Monte Carol method.

In fact, μ_{tw} and μ_{nw} can be easily linked to RP and Cs discussed in Section 3 with the following two formulas,

$$\mu_{tw} = RP = 1 - e^{-RP_0} \quad (10)$$

$$\mu_{nw} = \frac{Cs}{f_L \times N_b} \times N_{CM} \quad (11)$$

where N_b is the total number of time bins in the timing histogram ($N_b = 4000$ for $f_L = 1$ MHz) and N_{CM} is the number of time bins in a CM window

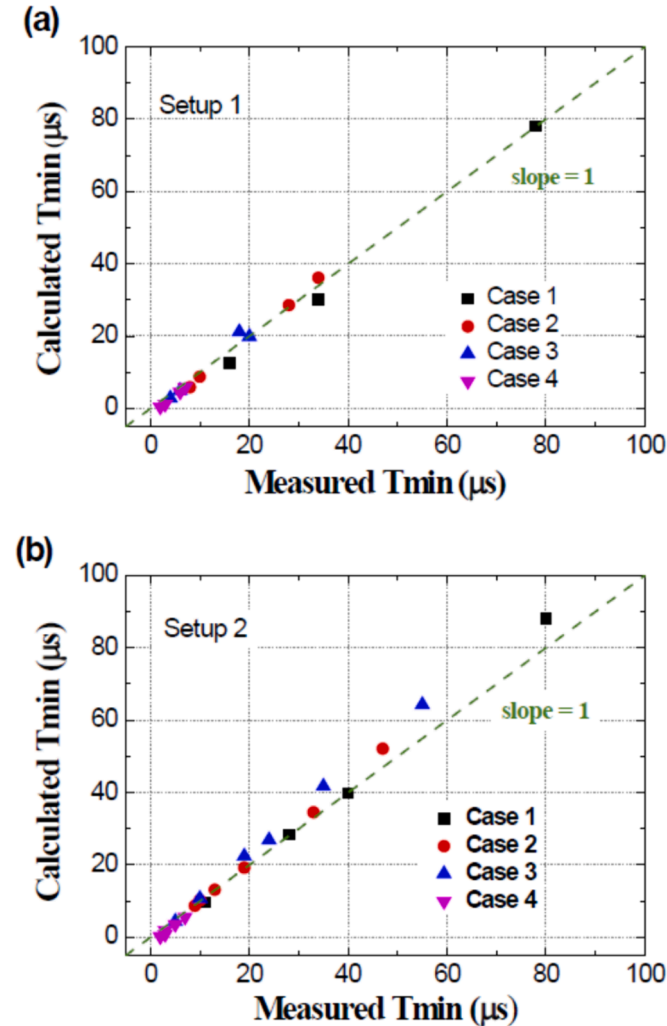


Fig. 7. Measured and calculated minimum ranging time $T_{\min}$ for setup 1 in (a) and setup 2 in (b) under the four experimental conditions cases 1–4.

($N_{CM}=16$ for 4 ns). Note that, in Eq. (10), the RP calculated from RP_0 is simply contributed by the laser source but μ_{tw} also includes the background noise counts. This may appear contradictory at first glance. Considering the deadtime of SPADs, the background counts coming just before laser photons' arrival reduce the laser counts in the target window but the background photons arriving at the target window also contribute to μ_{tw} . Therefore, μ_{tw} remain mostly unchanged and equal to RP under strong background sunlight as we have observed in our experiments (not shown here). With the relations in Eqs. (10) and (11), Link 2 in Fig. 1 has been established. Predicting $T_{\min}$ with given physical

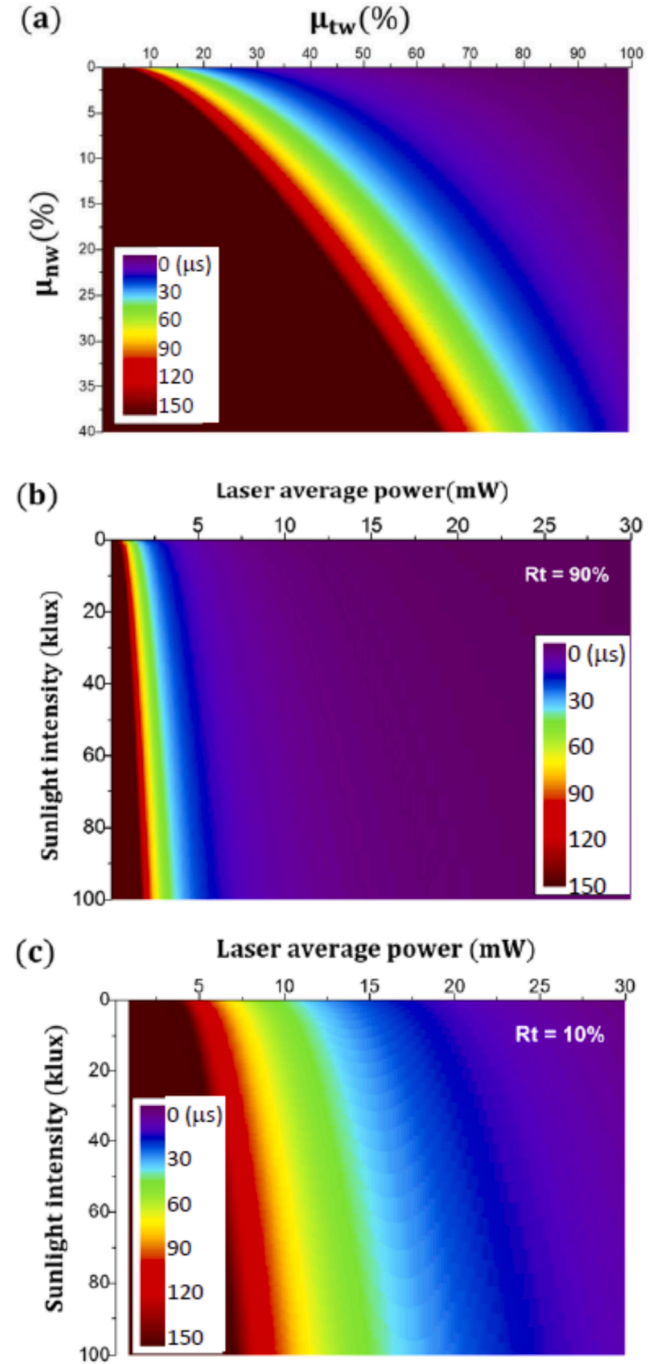


Fig. 8. (a) Minimum ranging time $T_{\min}$ as a function of counts in target window per laser pulse μ_{tw} and counts in noise window per laser pulse μ_{nw} . (b) (c) Minimum ranging time $T_{\min}$ as a function of sunlight intensity and laser average power for the target reflectivity of 90% in (b) and 10% in (c).

parameters of a LiDAR module now becomes straightforward. Fig. 8 shows minimum ranging time $T_{\min}$ as a function of μ_{TW} and μ_{NW} in (a) and of laser average power and sunlight intensity in (b) for $R_t = 90\%$ and in (c) for $R_t = 10\%$. The target distance D was set at 100 m and other parameters refer to setup 2 in Table 1.

The color plots in Fig. 8 are very useful as they tell us how to shorten the minimum ranging time under certain experiment conditions. For example, the $T_{\min}$ as a function of μ_{TW} and μ_{NW} in Fig. 8a indicates raising μ_{TW} is more effective than decreasing μ_{NW} . A nearly 100 % μ_{TW} could keep $T_{\min}$ under 10 μs for all background conditions. Comparing Fig. 8b and c also tells us the low reflectivity target at high background condition is the most difficult condition to be tackled with. More importantly, these relations give quantitative numbers instead of qualitative dependence between the physical parameters and $T_{\min}$, which would enable the engineering and optimizing of LiDAR architecture design.

$T_{\min}$ plays a deterministic role in a LiDAR module. For example, the frame rate can be calculated straight away with the given pixel number on each frame. Provided with the background sunlight and the target reflectance, one can also predict the longest detectable distance under the fixed frame rate and successful detection rate to choose the laser repetition rate. In addition, it is worth noting that the other key parameter, the ranging depth precision, can also be evaluated by knowing the minimum ranging time because, according to a recent paper [13], the ranging precision σ_p improves with the increasing integration time T_{int} roughly in a relation of $\sigma_p \propto T_{\text{int}}^{-1/2}$.

5. Conclusions

We have presented and demonstrated an analytical model for SPAD-based LiDAR system consisted of CMOS SPAD chips, pulsed semiconductor lasers, and a time-correlated photon-count card. The minimum ranging time, a core parameter for high-speed ToF ranging, has been calculated with our model from the physical parameters. The experimental verifications, including return photo-counts RP, sunlight background counts Cs, and minimum ranging time $T_{\min}$, have been carried out and the validity and effectiveness of our model has been demonstrated. Our work paves the way for the development of high-performance and low-cost LiDAR by providing a simple, analytical, and effective quantitative model for photon-counting ToF ranging.

CRediT authorship contribution statement

Po-Hsuan Chen: Methodology, Investigation, Data curation. **Chun-Hsien Liu:** Investigation. **Sheng-Di Lin:** Writing – review & editing, Writing – original draft, Project administration, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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